Sales Performance Analysis & Reporting System

SPAR — Project Report

End-to-End IT Project Management Portfolio Project

Prepared by	Snigdha Nath
Programme	MA in (IT) Project Management
Project ID	SPAR-1 | v1.0.0-MVP
Tools Used	Excel - Poer Point - Power Query - Jira - Confluence
Date	June 2026

1. Executive Summary

This report documents the end-to-end delivery of the Sales Performance Analysis and Reporting (SPAR) project.

The project simulates a real-world IT PM engagement where a retail sales team had no structured reporting system and operated on raw, inconsistent transaction data. As Project Manager, I was responsible for the full delivery lifecycle from data ingestion and cleaning, through analysis and dashboard development, to stakeholder reporting and project closure.

Project Objectives

- Clean and standardize a raw sales dataset using Power Query
- Analyze performance data across 5 key dimensions using Pivot Tables
- Deliver an interactive executive dashboard with dynamic slicers
- Manage end-to-end delivery across 3 Agile sprints in Jira
- Document the full project lifecycle in Confluence
- Present findings in a consulting-style stakeholder report

Headline Results

$366,751	**1,244**	**26%**	**August**
Total Revenue	**Total Orders**	**Return Rate**	**Peak Month**
FY2024 full year	Total units sold	Industry avg ~10%	$41,888 revenue

The project was delivered across 3 sprints from 3 June to 2 July 2026, with 11 of 14 work items completed and the epic at 85% progress as of reporting date.

2. Project Overview & Methodology

2.1 Business Problem

A retail sales team operated with raw, inconsistent FY2024 transaction data and had no mechanism for reporting or performance monitoring. Key business questions could not be answered:

- Which products and regions are driving revenue?
- Which sales representatives are underperforming?
- What is the return rate and what is causing it?
- Are there seasonal trends that require proactive planning?

The absence of a reporting system meant leadership was making decisions without data visibility

2.2 Project Approach

The project followed an Agile Scrum delivery model with 3 structured sprints, managed under a single epic in Jira. Each sprint had a clear focus area:

Sprint	Focus	Deliverables	Dates
Sprint 1	Data Preparation	Raw dataset sourced, loaded into Power Query, cleaned and transformed, data quality documented	3–10 Jun
Sprint 2	Analysis	5 pivot tables built, results validated, KPI cards defined, data mismatch bug resolved	10–17 Jun
Sprint 3	Reporting & Delivery	Executive dashboard built, dashboard reviewed, project documented and closed	17 Jun–2 Jul

2.3 Tools & Technologies

Tool	Category	Purpose in This Project
Microsoft Excel	Analysis & Reporting	Pivot tables, dashboard design, KPI cards, slicers, charts
Power Query	Data Engineering	Data cleaning, transformation, derived column creation
Jira (Scrum)	Project Management	Sprint planning, backlog management, issue tracking, epic management
Confluence	Documentation	Project charter, data dictionary, cleaning log, user guide, retrospective
PowerPoint	Communication	Consulting-style stakeholder presentation with findings and recommendations

3. Data Management

3.1 Raw Dataset Profile

The raw dataset was a simulated sales transaction file containing 305 rows and 9 columns. The dataset intentionally contained multiple data quality issues representative of real-world operational data:

Issue Type	Description
Blank rows	Fully empty rows scattered throughout the dataset
Duplicate Order IDs	Same Order ID appearing multiple times — inflating transaction counts
Inconsistent text casing	Product, Region and Sales Rep names in mixed formats e.g. 'laptop pro', 'LAPTOP PRO', 'Laptop Pro'
Mixed date formats	Four different date formats present: YYYY-MM-DD, DD/MM/YYYY, MM-DD-YYYY, DD-MMM-YYYY
Missing values	Some Unit Price and Total Amount cells blank or zero
Incorrect totals	Some Total Amount values did not match Quantity x Unit Price calculation

3.2 Power Query Cleaning Process

All data cleaning was performed in Microsoft Excel Power Query. The following steps were applied in sequence:

Step	Action	Rationale	Row Impact
1	Removed blank rows manually	Rows contained no usable data	305 → 301
2	Removed duplicate Order IDs	Duplicate transactions would inflate revenue and order metrics	301 → ~270
3	Standardised text columns	Inconsistent casing prevented accurate grouping in pivot tables	No row loss
4	Resolved mixed date formats	Four different date formats made date-based analysis unreliable	Some rows failed
5	Removed null date rows	Rows with unresolvable dates were unusable for time-based analysis	→ 240 final
6	Added Month Name and Quarter columns	Derived fields required for time-based pivot table analysis	No row loss
FINAL	**Clean dataset ready for analysis**	46 rows excluded (~15%) — decision documented and accepted	**259 rows**

3.3 Data Governance Decision

Data Loss Statement

The raw dataset of 305 rows was reduced to 240 clean rows, a data loss of approximately 15%. All excluded rows contained unrecoverable data quality issues (blank rows, duplicate IDs, or unresolvable date formats). This decision was made consciously, documented in the project Confluence log, and accepted by the project owner. The remaining 240 rows are complete and representative across all regions, products, months and sales representatives.

4. Analysis & Key Findings

Five pivot tables were built from the clean dataset to analyze performance across key business dimensions. Findings are presented below with supporting data and business interpretation.

4.1 Revenue by Month

Jan	Feb	Mar	Apr	May	Jun	...
$24,369	$8,907	$22,005	$23,723	$11,524	$30,341	→

Finding: August was the peak month at $41,888. February ($8,907) and May ($11,524) were significantly below the monthly average of $30,563 — indicating seasonal dips that could benefit from targeted promotional activity.

4.2 Revenue by Product

Product	Revenue	% of Total	Risk Flag
Laptop Pro	**$265,422**	**72%**	**HIGH RISK**
Monitor 24"	$37,455	10%	
Keyboard Mechanical	$18,510	5%	
Headset Pro	$16,547	5%	
USB Hub	$11,010	3%	
Webcam HD	$11,800	3%	
Wireless Mouse	$6,007	2%	

Finding: Laptop Pro generates 72% of total revenue, which is a critical concentration risk. If this product line faces supply disruption, pricing pressure or increased competition, the business is significantly exposed.

4.3 Revenue by Region

South	West	North	East
$122,091	$96,778	$89,975	**$55,907**

Finding: South leads at $122,091 while East significantly underperforms at $55,907 — a gap of 118%. This warrants investigation into East region’s headcount, territory coverage and pipeline health.

4.4 Sales Representative Performance

Sales Rep	Revenue	vs Average	Performance
Bob Smith	$88,864	+22%	**Top Performer**
Carol White	$83,865	+15%	Strong
Alice Johnson	$76,338	+5%	Average
Emma Davis	$68,028	-7%	Below Average
David Lee	$49,656	-32%	**Needs Support**

Finding: Bob Smith outperforms David Lee by 79%. The team average is approximately $73,377. David Lee is 32% below average — a gap significant enough to warrant a structured coaching or performance support intervention.

4.5 Order Status Breakdown

Delivered	Pending	Returned
124 orders — 41%	102 orders — 34%	**74 orders — 26%**

Finding: A 26% return rate is nearly three times the retail industry average of approximately 10%. This is the most operationally urgent finding in the dataset and requires immediate root cause investigation.

5. Business Recommendations

Based on the FY2024 data analysis, five prioritized recommendations are presented below. Each is linked to a specific finding, assigned an owner, and rated by urgency.

HIGH	**R1 — Reduce Revenue Concentration Risk** **Finding:** Laptop Pro generates 72% of total revenue ($265K of $367K) **Action:** Develop a product diversification strategy. Set revenue growth targets for underperforming product lines. Reduce Laptop Pro dependency to below 50% within 2 years.
HIGH	**R2 — Investigate and Reduce Return Rate** **Finding:** 26% return rate — nearly 3x the industry average of ~10% **Action:** Conduct root cause analysis on 74 returned orders. Identify if returns cluster by product, region or sales rep. Implement an order verification checklist at point of sale.
MED	**R3 — Address East Region Underperformance** **Finding:** East ($56K) trails South ($122K) by 118% **Action:** Review of the East region headcount, territory size and pipeline health. Consider mentoring program pairing East reps with South top performers.
MED	**R4 — Replicate Top Performer Best Practices** **Finding:** Bob Smith ($89K) outperforms David Lee ($50K) by 79% **Action:** Document Bob Smith's approach — product mix, engagement frequency, closing tactics. Build a replicable training program for the wider team.
LOW	**R5 — Address Seasonal Revenue Dips** **Finding:** February ($8,907) and May ($11,524) are significantly below the monthly average of $30,563 **Action:** Plan targeted promotional campaigns or sales incentive schemes for low revenue months. Align marketing budget to counterbalance seasonal trends proactively.

6. Project Management & Delivery

6.1 Jira Board Summary

Total Work Items	Done	In Progress	To Do
14	**11 (79%)**	**2 (14%)**	1 (7%)

The project was tracked using Jira Scrum under project key SPAR. A single epic — Sales Dashboard Delivery (SPAR-1) — contained all user stories and bugs across 3 sprints. The epic reached 85% completion as of reporting date, with final documentation (SPAR-12) remaining as the sole To Do item.

6.2 Issue Types & Classification

Issue Key	Summary	Type | Status
SPAR-3	Obtain raw sales dataset	Story | DONE
SPAR-4	Load data into Power Query	Story | DONE
SPAR-5	Clean and transform data	Story | DONE
SPAR-6	Document data quality findings	Story | DONE
SPAR-7	Build pivot tables for 5 KPIs	Story | DONE
SPAR-8	Validate pivot table results	Story | DONE
SPAR-9	Define dashboard KPI cards	Story | DONE
SPAR-17	Data Mismatch on Top Product KPI Card	Bug | DONE
SPAR-15	Chart Field Buttons Visible on Executive View	Bug | DONE
SPAR-16	Regional Slicer Broken on Product Revenue Chart	Bug | DONE
SPAR-10	Build executive dashboard	Story | DONE
SPAR-11	Conduct dashboard review	Story | IN PROGRESS
SPAR-12	Document and close project	Story | TO DO

6.3 Confluence Documentation

Document	Contents
Project Charter	Objective, scope, tools, timeline, stakeholders and success criteria
Data Dictionary	Definitions for all 11 columns in the clean dataset with data types and examples

Data Quality Log	6-step cleaning process with rationale and row impact at each stage
Dashboard User Guide	How to use KPI cards, interact with charts, and operate the 3 slicers
Project Retrospective	What went well, challenges, lessons learned and process improvements

7. Lessons Learned & Retrospective

A formal retrospective was conducted at the close of Sprint 3. Key observations are documented below across three dimensions — what went well, what was challenging, and what would be done differently.

What Went Well	What Was Challenging	What I Would Do Differently
• Power Query cleaned data efficiently across 6 structured steps • Dashboard was interactive and stakeholder-ready with 3 working slicers • Bugs were identified during review and resolved within the same sprint • Jira board clearly showed end-to-end project progress • Consulting-style presentation communicated findings clearly	• Mixed date formats in raw data caused errors requiring multiple resolution attempts • 15% data loss required a conscious governance decision • Story points were not assigned before sprint start — reports were empty • Dashboard bugs emerged during review — QA buffer was insufficient	• Validate and standardise data format at source before ingestion • Assign story points during sprint planning before sprint start • Build a dedicated QA sprint buffer into Sprint 3 • Link Jira issues to Confluence pages from Sprint 1 for better traceability

Key Takeaway

The most valuable moment in this project was not building the dashboard — it was deciding what to do with 46 rows of unrecoverable data. Documenting that decision transparently, explaining the rationale, and accepting accountability for the outcome is what separates a Project Manager from a technician. Data governance, stakeholder communication, and structured delivery are the core skills this project was designed to demonstrate — and they are exactly the skills required in a real IT PM role.

8. Conclusion

The SPAR project successfully demonstrated end-to-end IT project management delivery across all four planned tools — Microsoft Excel, Power Query, Jira, and Confluence — with a consulting-style stakeholder presentation as the final output.

Starting from a raw, messy dataset of 305 rows, the project delivered a clean, interactive executive dashboard surfacing five meaningful business insights and five prioritised recommendations for a retail sales business. The project was managed across three Agile sprints, fully documented in Confluence, and tracked in Jira under a structured epic and release framework.

This project demonstrates the following IT PM competencies:

- End-to-end delivery ownership — from problem definition to stakeholder communication
- Data literacy — understanding data quality, governance decisions, and their business impact
- Agile project management — sprint planning, backlog management, issue tracking in Jira
- Technical tool proficiency — Excel, Power Query, Pivot Tables, Slicers
- Documentation standards — structured Confluence pages across the full project lifecycle
- Consulting communication — findings and recommendations presented in Big 4 style

Snigdha Nath
MA Student — IT Project Management
June 2026